

# Task 1: Data Cleaning & Preprocessing

## Objective

Learn how to clean and prepare raw data for Machine Learning.

## Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

## Dataset

Titanic Dataset

## Steps Performed

1. Loaded the dataset using Pandas.
2. Explored dataset information and checked data types.
3. Identified missing values.
4. Filled missing values using median and mode.
5. Converted categorical features into numerical values using encoding.
6. Standardized numerical features.
7. Visualized outliers using boxplots.
8. Removed outliers using the IQR method.
9. Saved the cleaned dataset.

## Outcome

The dataset was successfully cleaned and prepared for Machine Learning model training.